MIL-E-1/1702A
6 DECEMBER 1985
SUPERSEDING
MIL-E-1/1702(NAVY)
4 February 1971

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 8745

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: UHF triode; conduction, convection, and forced-air cooled, ceramic and metal

See figure 1
Mounting position: Any

ABSOLUTE RATINGS:

Parameter: / Unit :	F1 MHz	Ef V Note 1	Eb Vdc	epy kv	Ec Vdc	Ib mAdc	ib a Note 2	ic a	tp µs	Du --
Maximum:										
Anode pulsed Osc. or Amp.:	3,000	6.0±5%	---	4.5	-150	---	1.5	0.7	250	0.005
Grid pulsed Osc. or Amp.:	3,000	6.0±5%	3500	---	-150	---	1.5	0.7	250	0.005
TEST CONDITIONS:	---	6.0	1000	---	Adj.	100	---	---	---	---

ABSOLUTE RATINGS:

Parameter: / Unit :	Pp W	Pg W Note 3	tk s (min)	TE °C Note 4	T Anode °C Note 4	Shank °C	Cooling --- Notes 4 and 5
Maximum:							
Anode pulsed Osc. or Amp.:	100	2.0	60	250	250		---
Grid pulsed Osc. or Amp.:	100	2.0	60	250	250		---
TEST CONDITIONS:	---	---	300	---	---		Note 6

GENERAL:

Qualification: Required

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:21Z
Check the source to verify that this is the current version before use.

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1	7					
1301	Heater current			If	0.90	1.05	A
1261	Electrode voltage (1) (grid)			Ec	-2.5	-7.5	Vdc
1211	Insulation of electrodes		Eb = Ek = 0; Ec = -500 Vdc	Rgk	50	---	Meg
1266	Total grid current			Ic	---	-8.0	uAdc
1231	Pulsing emission		Ef = 5.7 V; eb = ec = etd/is = 5.0 a; tp = 3 µs (max); prr = 600 (max)	etd	---	150	v
---	Video pulse operation	8	Ebb = 4,000 Vdc; Ec = -150 Vdc; egk/ik = 1.0 a (min); tp = 250 µs ±10%; prr = 20 ±10%; t = 60 seconds	---	---	---	---
---	High-voltage hold-off	9	Eb = 5,000 Vdc; Ec = -150 Vdc; t = 30 seconds	Ib	---	1.0	mAdc
	Quality conformance inspection, part 2	10					
1261	Electrode voltage (2) (grid)		Ec/Ib = 1.0 mAdc	Eco	---	-25.0	Vdc
1236	Power oscillation (pulse)	11	F = 3,000 MHz (min); epy = 3.5 kv; Ec = -1.5 Vdc (min); Rg/Ib = 7.5 mAdc; Ic = 4.5 mAdc (max); Ef = 5.8 V	Po	4.0	---	W (useful)
1331	Direct-interelectrode capacitance		Use fixture per Drawing 158-JAN; ground cathode connection	Cin Cgp Cout	5.60 1.85 ---	7.00 2.10 0.035	pF pF pF
---	Resonance test	12, 13	No voltages applied	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:21Z
Check the source to verify that this is the current version before use.

Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3	14, 15					
1031	Variable-frequency vibration	10, 16	F = 55 to 500 Hz; Acceleration = 10 G (peak); Ebb = 300 Vdc; Rp = 10,000 ohms; Ec/Ib = 10 mAdc	Ep	---	250	mVac
---	Torque	10, 17	No voltages	---	---	---	---
---	Torque-test end point						
1266	Total grid current			Ic	---	-10	µAdc
1042	Shock, specified pulse		Condition A; use fixture per Drawing 280-JAN; no voltages	---	---	---	---
---	Shock, specified pulse-test end point						
1266	Total grid current			Ic	---	-10	µAdc
1002	Barometric pressure, reduced	18	Press = 35 mmHg (max); Voltage = 1,800 Vac, 60 Hz; TA = 30°C±10°C	---	---	---	---
---	Life	19	Group S, monthly; E_f = 5.7 V; Filament standby	---	---	---	---
---	Life-test end point (1,000 hours)	19		ΔIb	---	25	%

NOTES:

1. The transit-time heating effect of the cathode may require compensation by a reduction in heater voltage after dynamic operation of the tube has started. The back heating is a function of frequency, grid current, grid bias, anode current, duty cycle, and circuit design and adjustment. There is an optimum heater voltage which will maintain the cathode at the correct operating temperature for a particular set of operating conditions. A maximum variation of ±5 percent from optimum is permitted. No reduction in heater voltage is required up to and including 500 MHz.

2. The regulation or series-anode-supply impedance, or both, shall limit the instantaneous peak current, with the tube considered as a short circuit, to a maximum of 10 times the specified maximum current rating.

3. The maximum instantaneous peak grid-cathode voltage shall be within the range of +250 to -750 volts.

4. Sufficient conduction, convection, and forced-air cooling shall be provided to limit the envelope and anode shank temperature to the specified value under all operating conditions. Reliability will be seriously impaired if this maximum is exceeded. Where emphasis is placed on long and reliable life, lower temperatures should be used.

5. At an anode dissipation of 100 watts and with an incoming air temperature of 25°C maximum at sea level, a minimum air flow of 12.5 cfm must be directed across the anode cooler, using the cowl as shown on Drawing 157-JAN, or equivalent.

Source: https://assist.dla.mil -- Downloaded: 202610-15T15:21Z
Check the source to verify that this is the current version before use.

NOTES: - Continued

6. Sufficient conduction, convection, and forced-air cooling may be used in all electrical tests involving application of heater voltage to maintain the anode shank and seal temperatures within the specified maximum value.

7. All tests listed under QCI, part 1, shall be performed at the conclusion of the holding period. The AQL (percent defective) for each test shall be 0.65, inspection level II.

8. Test in circuit on figure 2 or equivalent. There shall be no indication of any arc during the 60 second test period.

9. There shall be no indication of any arc during a test period of 60 seconds.

10. Other tube contact configurations may be used provided the tube contact area remains unchanged and the socket, jig, or cavity gives equal performance. Mounting of the socket, jig, or cavity may be at the option of the manufacturer.

11. The applied voltage pulse shape shall be measured with a noninductive resistor of 1,150 ohms ±2% inserted in place of the tube. The pulse shape shall be: tp = 0.4 µs maximum, and tf = 0.7 µs maximum. The pulse repetition rate (prr) shall be adjusted so that Du = 0.0025±5% with the above measured pulse length. Test in cavity in accordance with Drawing 279-JAN. The cavity shall be connected to a load with a VSWR of less than 1.5:1. The oscillator output coupling and the grid or cathode resistor may be adjusted for maximum power output.

12. Grid-anode resonance: Test in cavity in accordance with Drawing 278-JAN. Cavity shall resonate at 1354 ±2.0 MHz with tuning slug in accordance with Drawing 277-JAN at TA = 25°C±5°C.

    Grid-cathode resonance: Test in cavity in accordance with Drawing 283-JAN. Cavity shall resonate at 1719 ±2.0 MHz with tuning slug in accordance with Drawing 277-JAN at TA = 25°C±5°C.

    When plotted on graphs of resonant frequency versus grid-anode capacitance and resonant frequency versus grid-cathode capacitance, the TUT shall be represented by a point within the parallelogram whose four corners are located by the following points:

Points	Capacitance (pF)		Frequency (MHz)	
	C-gp	C-gk	F-gp	F-gk
1	1.85	5.6	1980	1790
2	1.85	5.6	2010	1820
3	2.10	7.0	1910	1740
4	2.10	7.0	1940	1770

13. The AQL (percent defective) shall be 6.5, inspection level S3.

14. When tube type 8745 is physically identical to tube type 7815 (except for the installation of the transverse anode cooler assembly), and has been made in the same production period as tubes which are being or have been tested to the applicable TSS for tube type 7815, then the one sample may represent both types insofar as the listed tests are identical.

15. These tests shall be performed during the initial production and once each succeeding 12-calendar month period in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes having an acceptance number of zero, and a second sample of three tubes having a combined acceptance number of one. In the event of failure, the test shall be made as a part of quality conformance inspection, part 2, inspection level S3, with an AQL of 6.5. The regular "12 calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:21Z
Check the source to verify that this is the current version before use.

NOTES: - Continued

16. The TUT shall be mounted in a socket in accordance with Drawing 276-JAN and vibrated with simple harmonic motion. The peak acceleration over the frequency range shall be within ±20 percent of the reference acceleration at 100 Hz. The frequency shall vary from 55 to 500 Hz and return to 55 Hz with approximately logarithmic progression and shall require 4 minutes minimum, 6 minutes maximum to traverse the range. Each tube shall be vibrated for 30 minutes in each position X and Z except that if the cumulative results of tests on 50 or more tubes of a construction show that more than 75 percent of the tubes have higher output voltages in one position, subsequent measurements need be taken only in the position giving the higher reading. The value of the alternating voltage, Ep, produced across the resistor Rp, as a result of vibration shall be measured with a suitable device. This device shall have an appropriate voltage range and shall have the ability to measure with an error of less than 10 percent the rms value of a sine wave of voltage at all frequencies from 20 to 20,000 Hz. The value of the vibrational output Ep shall not exceed the limit specified herein at any point in the sweep-frequency range during the last complete cycle of the cycling vibration.

17. The torque test shall be performed as follows:

    a. The tube shall be held securely at the cathode connection. A force of 5 pounds shall be applied to the heater cup without perceptible shock. This test may be made by applying the force at right angles to the inside of the cup at a point 7/64±1/64 inch from the cathode end of the tube. An approved equivalent method may be used. The heater cup shall not loosen or short circuit on the cathode connection. This part of the test shall not be required if the space between the heater cup and the cathode sleeve is completely filled with insulating material.

    b. A torque of 15 inch-pounds shall be applied between anode and cathode without shock.

    c. A torque of 40 inch-pounds shall be applied between anode and grid without shock.

18. Voltage shall be applied between the anode and grid. No other voltages shall be applied.

19. At zero hours, establish the drive conditions necessary to obtain 2.0 amperes peak anode current with an anode voltage of 1,000 Vdc and a bias voltage of -40 Vdc and Ef = 6.0 V. The pulse width of the modulator shall be 2 µs (minimum) and the duty cycle shall be 0.0025 maximum. With the drive level determined at zero hours, check the anode current at the end of life. The maximum allowable drop in anode current (Δib) is as specified.

20. Revision letters are not used to denote changes due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 202610-15T15:21Z
Check the source to verify that this is the current version before use.

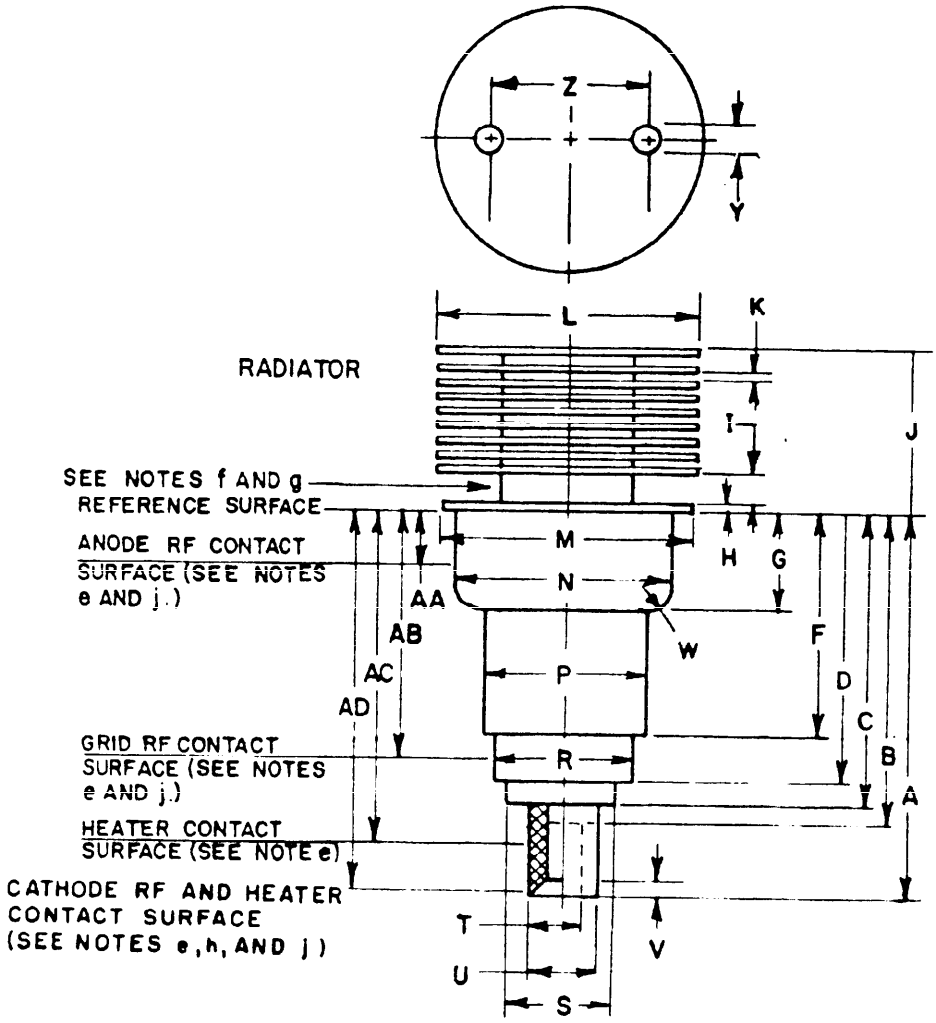


FIGURE 1. Outline drawing for electron tube type 8745.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:21Z
Check the source to verify that this is the current version before use.

Ltr	Dimensions in inches with metric equivalents (mm) in parenthesis		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			
A	1.815 (46.10)	1.875 (47.62)	
B	---	1.534 (38.96)	
C	---	1.475(37.46)	
D	1.289 (32.74)	1.329 (33.76)	
F	.970 (24.64)	1.010 (25.65)	
G	.462 (11.73)	.477 (12.12)	
J	.766 (19.46)	.826 (20.98)	
N	1.025 (26.04)	1.035 (26.29)	b,k
R	.655 (16.64)	.665 (16.89)	b,k
T	.213 (5.41)	.223 (5.66)	c,k
U	.315 (8.00)	.325 (8.26)	b,c,k
Quality conformance inspection, part 3			d
H	---	.040 (1.02)	
I	.125 (3.18)	.185 (4.70)	
K	.025 (.64)	.046 (1.17)	
L	1.234 (31.34)	1.264 (32.11)	
M	1.180 (29.97)	1.195 (30.35)	
P	.752 (19.10)	.792 (20.12)	
S	---	.545 (13.84)	
V	---	.086 (2.18)	
W	---	.100 (2.54)	
Y	.105 (2.67)	.145 (3.68)	
Z	.650 (16.51)	.850 (21.59)	
Electrode contact areas			a,m
AA	.035 (.89)	.361 (9.17)	b
AB	1.185 (30.10)	1.265 (32.13)	b
AC	1.534 (38.96)	1.728 (43.89)	c
AD	1.475 (37.46)	1.815 (46.10)	b,c

FIGURE 1. Outline drawing for electron tube type 8745 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:21Z
Check the source to verify that this is the current version before use.

NOTES:

a. Electrode contact surfaces and reference dimensions for eccentricity measurements.

b. The total indicated runout of the anode and grid contact surface with respect to the cathode contact surface shall not exceed .020 inch.

c. The total indicated runout of the cathode contact surface with respect to the heater contact surface shall not exceed .012 inch.

d. All dimensions shall be tested on 10 tubes per month when in continuous production. Failure of more than one tube to meet tolerances for any dimension shall cause that dimension to become, for all lots in process, part of QCI, part 2.

e. Silver plated 30 MSI minimum. Note d shall apply.

f. Plating not required over radiator of copper, aluminum, or approved equivalent.

g. This surface shall be used for measurement of anode shank temperature.

h. Inner edge of heater and outer edge of cathode rf connection shall be free from burrs and sharp edges.

j. Total indicated runout (TIR) of contact surfaces shall be gauged from centerline of reference and shall be as follows. Grid rf contact surface and reference dimension for eccentricity measurements.

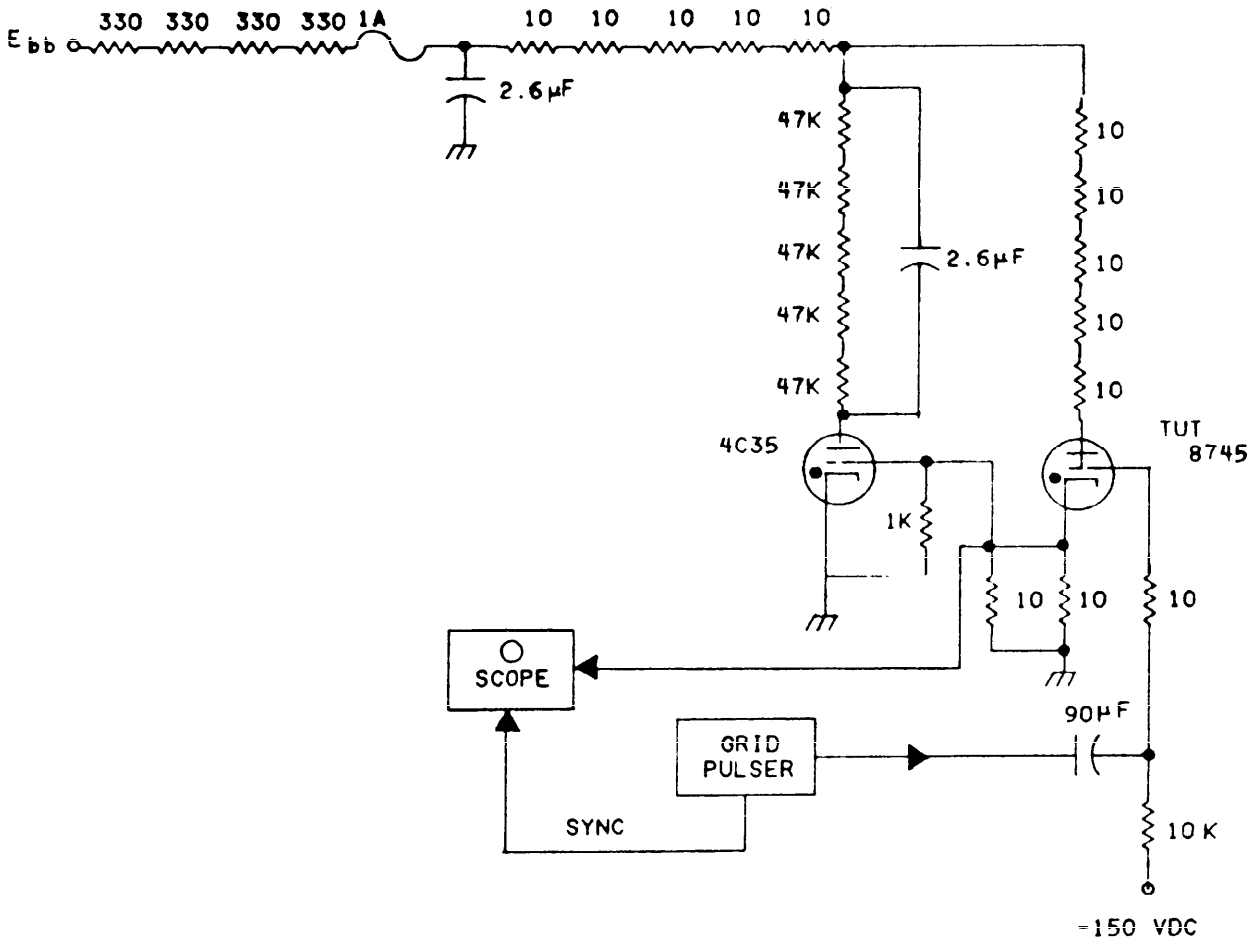
Contact surface	TIR, maximum	Reference
Anode	0.020	Cathode
Grid	0.020	Cathode
Heater	0.012	Cathode

k. Diameters N, R, T, and U shall apply throughout entire contact areas as defined by dimensions AA, AB, AC, and AD respectively.

m. Dimensions in electrode contact areas table are for socket design purposes and are not intended for inspection purposes.

FIGURE 1. Outline drawing for electron tube type 8745 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:21Z
Check the source to verify that this is the current version before use.



NOTE: All resistance in ohms, 2W.

FIGURE 2. Video pulse operation test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:21Z
Check the source to verify that this is the current version before use.

Custodians:
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5960-3362-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:21Z
Check the source to verify that this is the current version before use.